



# Modelling Financial Markets

## Chapter 3: Factor Models and Asset Pricing



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## Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ APT: Arbitrage Pricing Theory
- ▣ BAB: Betting Against Beta
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAGR: Compound Annual Growth Rate
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CMA: Conservative Minus Aggressive (investment factor)
- ▣ CML: Capital Market Line
- ▣ ETF: Exchange-Traded Fund
- ▣ FDR: False Discovery Rate
- ▣ FF3: Fama–French three-factor model
- ▣ FF5: Fama–French five-factor model
- ▣ FWER: Family-Wise Error Rate
- ▣ GLS: Generalised Least Squares
- ▣ GMM: Generalized Method of Moments
- ▣ GRS: Gibbons–Ross–Shanken (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HML: High Minus Low (value factor)
- ▣ IPCA: Instrumented Principal Component Analysis
- ▣ LASSO: Least Absolute Shrinkage and Selection Operator
- ▣ MOM: MOMentum factor
- ▣ NYSE: New York Stock Exchange
- ▣ OLS: Ordinary Least Squares
- ▣ OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))

## Acronyms used in this material (2/2)

- ▣ PC1: first Principal Component
- ▣ PC2: second Principal Component
- ▣ PC3: third Principal Component
- ▣ PCA: Principal Component Analysis
- ▣ PRDS: Positive Regression Dependence on a Subset (of p-values)
- ▣ RF: Risk-Free rate
- ▣ RMSE: Root Mean Squared Error
- ▣ RMW: Robust Minus Weak (profitability factor)
- ▣ ROE: Return On Equity
- ▣ RP-PCA: Risk-Premium Principal Component Analysis
- ▣ SDF: Stochastic Discount Factor
- ▣ SMB: Small Minus Big (size factor)
- ▣ SML: Security Market Line
- ▣ UMD: Up Minus Down (momentum factor)
- ▣ US: United States

## Chapter Route and Outcomes

### Four questions

- ▣ What explains expected returns? The CAPM and multifactor models
- ▣ How are exposures estimated? Betas, their uncertainty and shrinkage
- ▣ How are pricing restrictions tested? GRS, Fama–MacBeth, model comparison
- ▣ Do the findings survive? Multiple testing, latent factors, replication

Part I: Sections 1–5; Part II: Sections 6–13; technical detail in the Appendix

### After this chapter you can

- ▣ estimate a beta with HAC errors and shrink it
- ▣ test the CAPM jointly and estimate risk premia with robust errors
- ▣ judge a new factor against multiple-testing hurdles and replication evidence

## What Is a Factor Model?

- A **factor model** explains the return of asset  $i$  by a few common variables  $f_{1,t}, \dots, f_{K,t}$ , the **factors**

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,1}f_{1,t} + \dots + \beta_{i,K}f_{K,t} + \varepsilon_{i,t}$$

- $\beta_{i,k}$ : **factor loading** (exposure) of asset  $i$  to factor  $k$ ;  $\varepsilon_{i,t}$ : asset-specific (idiosyncratic) shock
- Two uses in this chapter
  - ▶ Explain an asset: the technology ETF XLK moves 1.28 times the US market (Section 3)
  - ▶ Evaluate a fund: is its  $\alpha_i$  positive once its exposures are priced (Sections 4 and 10)?
- Also: cost of capital, risk management ( $N$  assets need  $N(N+1)/2$  covariances,  $K$  factors far fewer), portfolio construction (Campbell, Lo & MacKinlay (1997), ch. 5–6)
- Key message: expected returns should be explained by **exposures to priced risks**, not by the name of the asset

## Data and Conventions

| Series                          | Content                                         | Period    |
|---------------------------------|-------------------------------------------------|-----------|
| Fama–French factors             | market, size, value, profitability, investment  | 1963–2026 |
| Momentum factor                 | winners minus losers                            | 1963–2026 |
| 25 size $\times$ B/M portfolios | US stocks, two sorts                            | 1963–2026 |
| US sector ETFs                  | S&P 500 sectors: 9 from 1999, 11 from 2018      | 1999–2026 |
| Factor ETFs                     | momentum, value, quality, min. volatility, size | 2013–2026 |
| BVB blue chips, BET             | 11 stocks and their index                       | 2015–2026 |

- ▣ Factors and portfolios: [Kenneth French Data Library](#), in percent (divide by 100); ETFs and stocks: daily adjusted closing prices (BET: closing values), to 18 Sep 2026
- ▣ **Excess return**  $R_{i,t}^e = R_{i,t} - R_{f,t}$ ;  $R_{f,t}$ : one-month Treasury bill rate (RF in the French data)
- ▣ **B/M**: book value of equity over its market value; **BET**: the benchmark index of the BVB
- ▣ Simple returns for regressions on the French factors; log returns for rolling, BVB and PCA analyses
- ▣ BVB: daily  $R_{f,t}$  treated as constant; it cancels in covariances, so betas are unchanged
- ▣ ETFs and stocks: dividend- and split-adjusted prices; indices: closing values

## Mean-Variance and the Tangency Portfolio

- **Markowitz (1952)**: with weights  $w$ , expected excess returns  $\mu$  and covariance  $\Sigma$ , the frontier solves

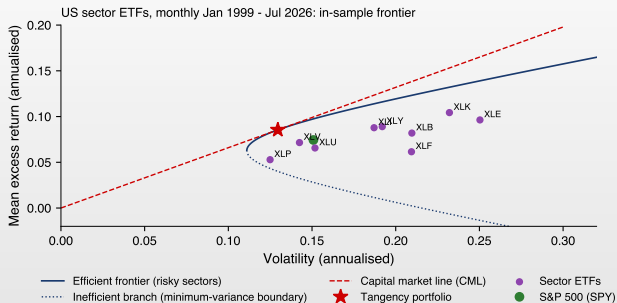
$$\min_w w^\top \Sigma w \quad \text{s.t.} \quad w^\top \mu = \mu^*, \quad w^\top \mathbf{1} = 1$$

- **Sharpe ratio**  $SR_p = \mathbb{E}[R_p - R_f] / \sigma_p$ ; with a risk-free asset the best risky portfolio is the **tangency portfolio**

$$w_{\text{tan}} = \Sigma^{-1} \mu / \mathbf{1}^\top \Sigma^{-1} \mu$$

- **CML**: mixes of  $R_f$  and the tangency portfolio,  $\mu_p = SR_{\text{tan}} \sigma_p$ ; negative weights are **short positions**
- From the tangency portfolio to the beta pricing equation: Seminar 3, problem A1; estimation error and shrinkage: Chapter 4

## The Efficient Frontier of US Sectors



- Monthly excess returns of 9 sector ETFs, January 1999 – July 2026 ( $T = 331$  months)
- Tangency portfolio: mean 8.55%, volatility 12.96%, Sharpe ratio 0.66
- SPY (the S&P 500 ETF, the market proxy): mean 7.44%, volatility 15.08%, Sharpe ratio 0.49: here the market is **not** on the frontier, as the CAPM would require
- Tangency weights:  $-0.37$  in financials,  $-0.21$  in materials,  $+0.37$  in health care
- The frontier is **in-sample**; estimated means are very noisy (Chapter 4)

 MFM\_ch3\_frontier

## The CAPM: Sharpe, Lintner, Black



William F. Sharpe, Nobel Prize in  
Economics 1990

Photo: Larry D. Moore (2007); CC BY  
4.0; Wikimedia Commons

- **CAPM**: Capital Asset Pricing Model, [Sharpe \(1964\)](#) and [Lintner \(1965\)](#)
- If every investor holds a mean–variance efficient portfolio, all hold the same tangency portfolio, which must then be the **market portfolio**
- Expected excess return is proportional to **beta**

$$\mathbb{E}[R_i] - R_f = \beta_i (\mathbb{E}[R_m] - R_f), \quad \beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$$

- $\mathbb{E}[R_m] - R_f$ : **market risk premium**;  $\beta_i$ : sensitivity of asset  $i$  to the market
- [Black \(1972\)](#): a version without a risk-free asset, the zero-beta CAPM (Section 4)
- Sharpe shared the 1990 Nobel Prize with Markowitz and Miller

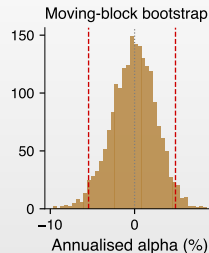
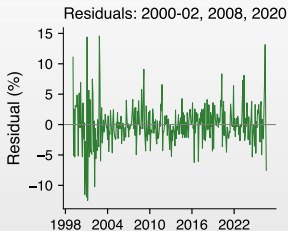
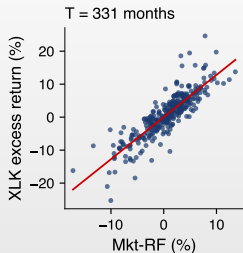
## What the CAPM Implies

- Main assumptions
  - ▶ Mean–variance investors with identical beliefs, one period
  - ▶ Unlimited borrowing and lending at  $R_f$ ; no taxes or trading costs
- Only beta is priced: no other characteristic should predict returns; idiosyncratic risk earns nothing
- **SML** (Security Market Line): every asset lies on the line (beta, mean) with slope equal to the premium; the CML contains only efficient portfolios. The SML is the testable statement (Section 4)
- Cost of capital from the CAPM needs an estimated beta and an estimated premium: the error of the premium dominates
- Which portfolio is “the market”? The answer decides what a test of the CAPM can establish (Roll’s critique, Section 4)

## The Market Model, HAC Errors and Estimation Choices

- ▣ Market model  $R_{i,t}^e = \alpha_i + \beta_i R_{m,t}^e + \varepsilon_{i,t}$ ; **OLS** (Ordinary Least Squares):  $\hat{\beta}_i = \widehat{\text{Cov}}(R_i^e, R_m^e) / \widehat{\text{Var}}(R_m^e)$
- ▣ Variance split  $\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma_{\varepsilon,i}^2$ : systematic (priced under the CAPM) plus idiosyncratic (diversifiable);  $R^2$  is the systematic share
- ▣ Volatility clustering (Chapter 1) makes residuals heteroskedastic: use **HAC** errors, [Newey & West \(1987\)](#), with Bartlett weights and  $L = \lfloor 4(T/100)^{2/9} \rfloor$  lags ( $T = 331$ :  $L = 5$ ;  $T = 757$ :  $L = 6$ ); the point estimates are those of OLS
- ▣ Choices to report with every beta
  - ▶ Frequency and window: 3–5 years of monthly or 1–2 years of daily data; daily data suffer from asynchronous trading
  - ▶ Market proxy: SPY (S&P 500) for US assets, BET for Romanian stocks
  - ▶ Data cleaning: an unadjusted corporate event creates a fake return (DIGI, May 2017: a daily log return of  $-6.6$ )

## Worked Example: Beta of the Technology ETF

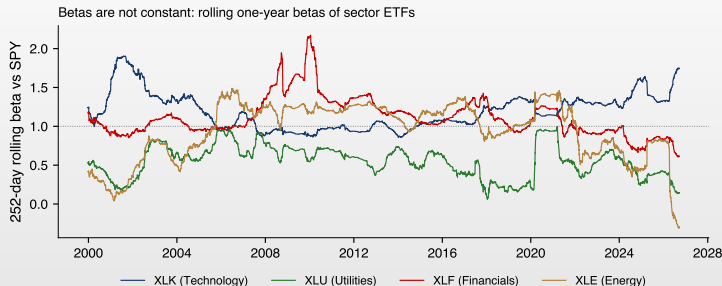


- Monthly excess returns
- OLS: beta = 1.277
- OLS residual
- Bootstrap alpha (2000 draws)
- - - 95% percentile interval

|                 | Estimate | Classical | White | Newey–West |
|-----------------|----------|-----------|-------|------------|
| $\alpha$ (ann.) | 0.33%    | 2.27%     | 2.32% | 2.45%      |
| $\beta$         | 1.277    | 0.041     | 0.054 | 0.082      |

- XLK on Mkt-RF, monthly 1999–2026,  $T = 331$ ,  $R^2 = 0.742$
- Robust s.e. of  $\beta$ : twice the classical one; heteroskedasticity explains part (White), serial correlation of  $f_t \varepsilon_t$  the rest
- $t(\alpha) = 0.33/2.45 = 0.13$ ; bootstrap interval  $[-5.5\%, 4.9\%]$

## Betas Are Not Constant



- OLS beta on the last 252 daily log returns, recomputed every day
- Financials (XLF): from 0.60 to 2.17; the peak follows the 2008–2009 banking crisis
- Technology (XLK): from 0.86 to 1.90; high in the dot-com years, 1.75 at the end
- Energy (XLE): from  $-0.31$  to 1.49; negative at the end, when oil moved against the market
- Utilities (XLU): from 0.06 to 1.00, the classic defensive sector
- A full-sample beta hides these regimes; conditional models: Chapters 5–6

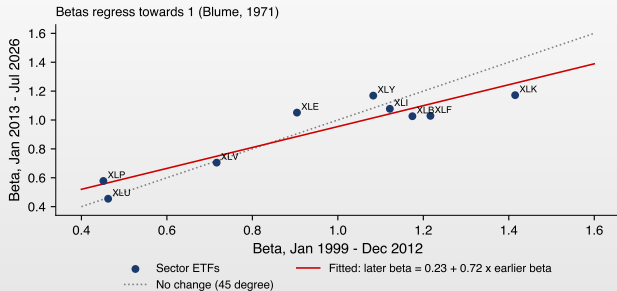
## Shrinking Betas: Blume and Vasicek

- ▣ **Regression to the mean:** extreme estimated betas are partly noise and move towards 1 later
- ▣ **Blume (1971):** regress later betas on earlier betas; the practitioner rule is  $\beta^{\text{Blume}} = 0.33 + 0.67 \hat{\beta}$
- ▣ **Vasicek (1973):** Bayesian average of the estimate and the cross-sectional prior mean  $\bar{\beta}$

$$\beta_i^V = w_i \hat{\beta}_i + (1 - w_i) \bar{\beta}, \quad w_i = \frac{\sigma_{\beta}^2}{\sigma_{\beta}^2 + \text{se}(\hat{\beta}_i)^2}$$

- ▣  $\sigma_{\beta}^2$  is the variance of the **true** betas:  $\text{Var}_{cs}(\hat{\beta}) = \sigma_{\beta}^2 + \overline{\text{se}^2}$ , so the empirical-Bayes prior is  $\sigma_{\beta}^2 = \text{Var}_{cs}(\hat{\beta}) - \overline{\text{se}^2}$  (posterior derivation: Seminar 3, problem A3)
- ▣ XLK, monthly 1999–2012, 9 sectors:  $\hat{\beta} = 1.415$  (se 0.139),  $\bar{\beta} = 0.949$ ,  $\sigma_{\beta}^2 = 0.116 - 0.008 = 0.108$ ,  $w = 0.849$   
 ►  $\beta^V = 0.849 \times 1.415 + 0.151 \times 0.949 = 1.344$ ; Blume 1.278; beta estimated on 2013–2026: 1.172

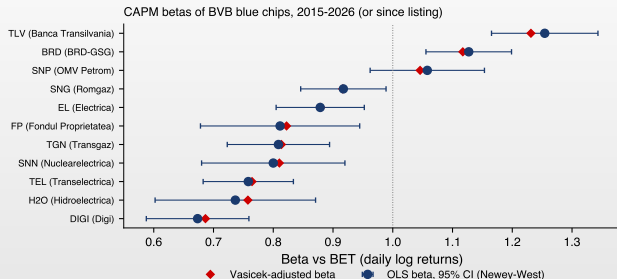
## Do Betas Regress Towards One?



- Monthly betas of 9 sector ETFs on SPY: 1999–2012 (horizontal) vs 2013–2026 (vertical)
- Fitted line  $\hat{\beta}_{\text{later}} = 0.23 + 0.72 \hat{\beta}_{\text{earlier}}$ , very close to Blume's  $0.33 + 0.67 \hat{\beta}$
- RMSE of the forecast of later betas,  $\sqrt{\frac{1}{N} \sum_i (\text{forecast}_i - \hat{\beta}_{i,\text{later}})^2}$ : raw 0.135, Blume 0.109, Vasicek 0.114
- Shrinkage cuts the forecast error by 15–20%; with 9 sectors the lesson is qualitative

 MFM\_ch3\_beta\_shrink

## CAPM Betas on the Bucharest Stock Exchange



- Daily log returns on BET, from 6 January 2015 or the listing date (Digi, DIGI, 2017; Hidroelectrica, H2O, 2023); days with  $|r_t| > 0.5$  removed (one for DIGI, two for Fondul Proprietatea, FP)
- Banks are the most market-sensitive: Banca Transilvania (TLV) 1.25, BRD 1.13; OMV Petrom 1.06; Transelectrica 0.76, Digi 0.67
- $R^2$ : 0.59 for TLV, only 0.21 for DIGI; large index weights raise  $R^2$  mechanically
- Vasicek changes little: the daily estimates are precise (se  $\approx 0.04$ – $0.07$ ); stale prices bias daily betas of less liquid stocks (Seminar 3, problem B4)

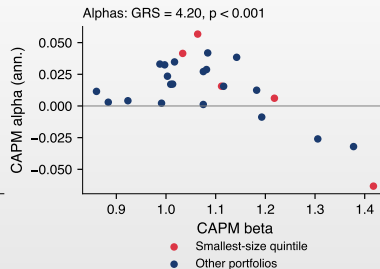
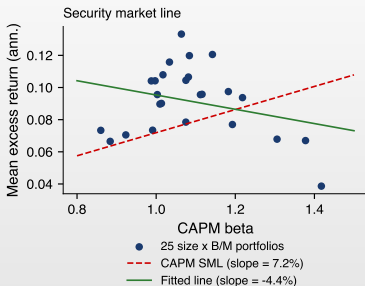
## From One Alpha to the GRS Test

- ▣ **Jensen's alpha** (Jensen (1968)) is the market-model intercept  $\alpha_i$ ; the CAPM implies  $\alpha_i = 0$  for every asset
- ▣ One  $t$ -test per asset (XLK:  $t = 0.13$ ) inflates false rejections across many assets (Section 8): **GRS**, Gibbons, Ross & Shanken (1989), tests  $H_0 : \alpha_1 = \dots = \alpha_N = 0$  jointly

$$W = \frac{T - N - 1}{N} \frac{\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \hat{\mu}_m^2 / \hat{\sigma}_m^2} \sim F(N, T - N - 1)$$

- ▣  $\hat{\Sigma}$ : maximum-likelihood residual covariance (divided by  $T$ );  $\hat{\mu}_m / \hat{\sigma}_m$ : Sharpe ratio of the market
- ▣ Exact, conditional on the factors, under i.i.d. multivariate Normal errors independent of the factors, with zero mean and constant nonsingular covariance;  $F$ : Fisher distribution
- ▣ Large  $W$ : the market is far from the mean–variance frontier of the test assets

## The Security Market Line in the Data



- Test assets: US stocks sorted each June into 5 size and 5 B/M quintiles, 25 value-weighted portfolios (Fama & French (1993)), July 1963 – July 2026,  $T = 757$
- The CAPM line has slope 7.2% a year; the fitted line has slope -4.4%: higher beta, **lower** average return

- Betas span only 0.86–1.42, average excess returns 3.9%–13.3%
- Small-value alpha +5.7%, small-growth -6.3% a year; 10 of 25 alphas have  $|t| > 1.96$
- GRS = 4.20,  $p$ -value  $8 \times 10^{-11}$ : the CAPM is rejected

## GRS as a Sharpe-Ratio Test

- Identity (Gibbons, Ross & Shanken (1989)): with  $\hat{\Sigma}$  and the moments estimated by maximum likelihood

$$\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha} = \widehat{SR}^2(f, R) - \widehat{SR}^2(f)$$

- ▶  $\widehat{SR}^2(f, R)$ : squared Sharpe ratio of the tangency portfolio of the factor and the  $N$  test assets

- So GRS measures the Sharpe-ratio gain from the test assets

$$W = \frac{T - N - 1}{N} \cdot \frac{\widehat{SR}^2(f, R) - \widehat{SR}^2(f)}{1 + \widehat{SR}^2(f)}$$

- 25 portfolios, monthly Sharpe ratios: market 0.134 (0.47 a year); market plus portfolios 0.405 (1.40 a year)
- $W = 29.24 \times (0.405^2 - 0.134^2)/(1 + 0.134^2) = 4.20$ ; the gain expected under  $H_0$  is far smaller (Appendix)

## When the GRS Assumptions Fail

- ▣ **Heavy tails and volatility clustering** (Chapter 1): the  $F$  distribution is no longer exact
  - ▶ Residual bootstrap under  $H_0$ : resample whole rows of residuals, rebuild returns with  $\alpha = 0$  (Seminar 3, problem B11); robust to non-Normality only if the errors are i.i.d. and independent of the factors
  - ▶ GMM Wald test: moments  $\varepsilon_t \otimes (1, f_t)$ ,  $W_{\text{GMM}} = \hat{\alpha}^\top \hat{V}_\alpha^{-1} \hat{\alpha} \rightarrow \chi_N^2$ , with  $\hat{V}_\alpha$  the HAC estimate of  $\text{Var}(\hat{\alpha})$ : robust to heteroskedasticity and autocorrelation
- ▣ **Rank condition**:  $\hat{\Sigma}$  has rank at most  $T - 2$ , so GRS is not computable when  $N \geq T - 1$ , and its power falls well before that point
- ▣ Tests built for many assets: Appendix

## A Flat SML: Black and Betting Against Beta



Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia  
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- **Black (1972)**: if borrowing is restricted, investors who want more risk buy high-beta assets instead of using leverage; high-beta assets become expensive and earn less
- Prediction: the SML is **flatter** than the CAPM line and starts at a **zero-beta** rate above  $R_f$
- **Frazzini & Pedersen (2014)** turn this into a strategy, **BAB** (Betting Against Beta)

$$r_t^{\text{BAB}} = \frac{R_{L,t} - R_{f,t}}{\beta_L} - \frac{R_{H,t} - R_{f,t}}{\beta_H}$$

- $L$ : low-beta portfolio leveraged to beta 1;  $H$ : high-beta portfolio de-leveraged to beta 1; BAB has beta zero by construction and a positive average return in their data
- An unscaled low-minus-high-beta spread (Seminar 3, problem C1) is not beta-neutral: its mean mixes the premium with a negative market exposure

## What a CAPM Test Establishes

- ▣ Roll (1977): the CAPM is a statement about the **true** market portfolio of all wealth; stock indices omit bonds, real estate, private firms and human capital
- ▣ Any efficient portfolio prices all assets exactly by beta: with  $R_f$  on the CML, with a zero-beta rate (Black (1972)) on the risky-asset frontier (Seminar 3, problem A1)
  - ▶ A CAPM test is a joint test of the model and of the efficiency of the proxy
- ▣ **Test assets matter**: 9 sector ETFs, January 1999 – July 2026 ( $T = 331$ ):  $GRS = 0.53$ ,  $p = 0.85$ , not rejected
  - ▶ On the same 331 months the 25 size  $\times$  B/M portfolios still reject:  $GRS = 2.91$ ,  $p = 9 \times 10^{-6}$
  - ▶ Sectors mix small and big, value and growth: they do not spread the characteristics that matter
- ▣ Failing to reject is weak evidence; evaluate models by how well they price assets with dispersion in expected returns

## Arbitrage Pricing Theory

- ▣ **APT**: Arbitrage Pricing Theory, [Ross \(1976\)](#)
- ▣ Returns follow a  $K$ -factor model with small idiosyncratic risks

$$R_{i,t} = \mathbb{E}[R_i] + \beta_i^\top f_t + \varepsilon_{i,t}, \quad \mathbb{E}[f_t] = 0$$

- ▣ **No arbitrage** (no riskless profit without investment) implies, approximately

$$\mathbb{E}[R_i] - R_f = \beta_i^\top \lambda$$

- ▣  $\lambda$ : the **risk premia**, the extra expected return per unit of each exposure
- ▣ The theory does not say **which** factors: they must be found in the data

## Building Factors: Long–Short Portfolios

size: NYSE median; B/M: 30% / 40% / 30%

|       | Low<br>(growth) | Neutral | High<br>(value) |
|-------|-----------------|---------|-----------------|
| Small | $S_L$           | $S_N$   | $S_H$           |
| Big   | $B_L$           | $B_N$   | $B_H$           |

$$\text{SMB} = \frac{1}{3}(S_L + S_N + S_H) - \frac{1}{3}(B_L + B_N + B_H)$$

$$\text{HML} = \frac{1}{2}(S_H + B_H) - \frac{1}{2}(S_L + B_L)$$

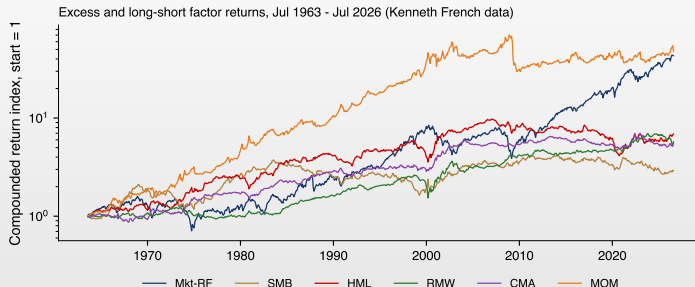
- A **characteristic-sorted factor** buys stocks with a high value of a characteristic and sells those with a low value
- Zero net investment: the return is a pure spread,  
 $f_t = R_{\text{high},t} - R_{\text{low},t}$
- Fama–French  $2 \times 3$  sort each June, accounting data of the previous year, value-weighted portfolios
  - ▶ **SMB** (Small Minus Big): small minus big stocks
  - ▶ **HML** (High Minus Low): high B/M (value) minus low B/M (growth)
- The same construction on profitability and investment gives RMW and CMA (next slide)

## The Standard Factor Models

| Model            | Factors                                    | Reference                                 | Motivation                                     |
|------------------|--------------------------------------------|-------------------------------------------|------------------------------------------------|
| CAPM             | Mkt-RF                                     | Sharpe (1964)                             | equilibrium                                    |
| FF3              | Mkt-RF, SMB, HML                           | Fama & French (1993)                      | size and B/M effects (Fama & French (1992))    |
| Carhart q-factor | FF3 + MOM<br>market, size, investment, ROE | Carhart (1997)<br>Hou, Xue & Zhang (2015) | fund evaluation<br>firms' investment decisions |
| FF5              | Mkt-RF, SMB, HML, RMW, CMA                 | Fama & French (2015)                      | dividend-discount model                        |

- FF3:  $R_{i,t}^e = \alpha_i + b_i \text{MKT}_t + s_i \text{SMB}_t + h_i \text{HML}_t + \varepsilon_{i,t}$ ; a good model leaves time-series alphas close to zero (GRS not rejected)
- **RMW** (Robust Minus Weak): high minus low operating profitability; **CMA** (Conservative Minus Aggressive): low minus high asset growth; ROE: Return On Equity
- **MOM** (UMD, Up Minus Down): high minus low return from month  $t - 12$  to  $t - 2$ ; past winners keep winning over 3–12 months (Jegadeesh & Titman (1993))
- FF3 uses its own SMB ( $2 \times 3$  sorts on size and B/M); the FF5 SMB averages size spreads from three sorts, so FF3 is not nested in FF5

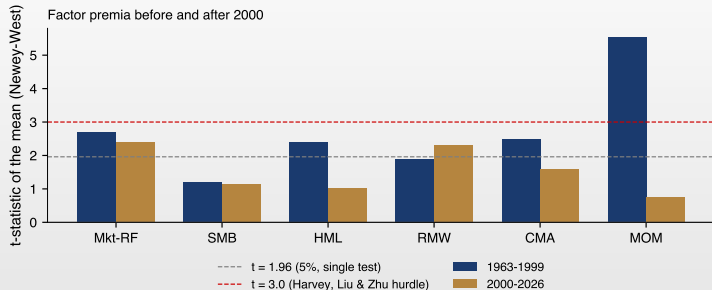
## Six Factors over Sixty-Three Years



- Compounded monthly factor returns from 1, log scale: excess or long-short returns, so no interest on capital or collateral; not the wealth of a funded investment
- July 1963 – July 2026: Mkt-RF 43.2, MOM 48.0, HML 6.9, RMW 5.7, CMA 5.5, SMB 2.9
- Momentum peaked before 2009 and lost heavily in the 2009 rebound: a **momentum crash**
- Value was flat or losing from about 2007 to 2020
- Average annual returns: Mkt-RF 7.2%, MOM 7.3%, HML 3.6%, RMW 3.1%, CMA 3.0%, SMB 2.3%

 MFM\_ch3\_factor\_cum

## Do the Premia Survive?



- $t$ -statistic of the mean monthly factor return (Newey–West), 1963–1999 vs 2000–2026
- Momentum:  $t = 5.53$  before 2000, 0.75 after; mean 10.7% vs 2.5% a year
- Value (HML):  $t = 2.39$  then 1.02; mean 4.2% vs 2.8%
- Profitability (RMW) is the exception:  $t = 1.90$  then 2.31; the market premium holds in both halves ( $t = 2.70$  and 2.39)
- A calendar split is not a publication event: dating the decay needs the design of Section 12

## The Two Passes



Eugene Fama, Morgan Stanley–AFA

Award, 2008

Photo: TonyTheTiger (2008); CC

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- ▣ Fama & MacBeth (1973) estimate the risk premia  $\lambda$  of  $\mathbb{E}[R_i^e] = \lambda_0 + \beta_i^\top \lambda$  (Fama: Nobel Prize 2013)
- ▣ **Pass 1** (time series, one regression per asset): returns  $T \times N$ , factors  $T \times K \Rightarrow$  betas  $\hat{B}$ ,  $N \times K$

$$R_{i,t}^e = a_i + \beta_i^\top f_t + \varepsilon_{i,t}$$

- ▣ **Pass 2** (cross section, one regression per month):  $N$  returns on  $X = [1, \hat{B}]$

$$R_{i,t}^e = \lambda_{0,t} + \hat{\beta}_i^\top \lambda_t + u_{i,t}, \quad \hat{\lambda} = \frac{1}{T} \sum_t \hat{\lambda}_t$$

- ▣  $\lambda_0$ : pricing error common to all assets, zero if the model is right
- ▣ Allows factors that are **not** traded portfolios, and firm characteristics as regressors

## Standard Errors for Risk Premia

- ▣ **Fama–MacBeth**: treat the monthly  $\hat{\lambda}_{k,t}$  as a sample of size  $T$ ,  $se_{FM}(\hat{\lambda}_k) = s(\hat{\lambda}_{k,t})/\sqrt{T}$ ; robust to cross-sectional correlation of residuals; if  $\hat{\lambda}_{k,t}$  is autocorrelated, use Newey–West on this series
- ▣ **Errors in variables**: pass 2 uses  $\hat{\beta}$ , not  $\beta$ , so  $se_{FM}$  is too small; [Shanken \(1992\)](#) scales the non-factor part by  $(1 + c)$

$$\text{Var}_{Sh}(\hat{\lambda}) = (1 + c) \left[ \text{Var}_{FM}(\hat{\lambda}) - \frac{1}{T} \Sigma_f \right] + \frac{1}{T} \Sigma_f, \quad c = \hat{\lambda}^\top \Sigma_f^{-1} \hat{\lambda}$$

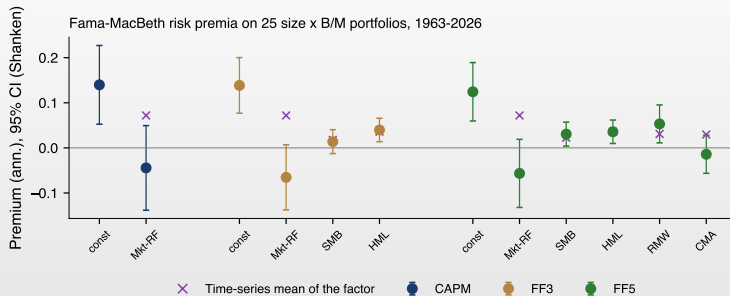
- ▶  $\Sigma_f$ : factor covariance;  $c$ : squared Sharpe ratio of the premia per period, 0.03 for FF3 here, larger for some non-traded factors
- ▣ **Misspecification**: all three assume a correct model, pricing errors  $e = \mu - X\gamma = 0$ ; when GRS rejects, the noise in  $\hat{B}$  multiplies  $e$  and adds variance ([Kan, Robotti & Shanken \(2013\)](#); GMM form in the Appendix)

## Worked Example: Four Standard Errors for the Value Premium

| FF3, HML            | $\hat{\lambda}$ | s.e.  | $t$  | Significant at 5%? |
|---------------------|-----------------|-------|------|--------------------|
| Fama–MacBeth        | 3.97%           | 1.33% | 2.99 | yes                |
| Newey–West (6 lags) | 3.97%           | 1.62% | 2.45 | yes                |
| Shanken             | 3.97%           | 1.33% | 2.99 | yes                |
| Kan–Robotti–Shanken | 3.97%           | 1.32% | 3.00 | yes                |

- 25 size  $\times$  B/M portfolios, 1963–2026; Shanken barely moves the error ( $c = 0.03$  per month)
- Autocorrelation of the monthly slopes  $\hat{\lambda}_t$  matters most here: Newey–West, 6 lags by the rule for  $T = 757$
- FF5: the misspecification-robust s.e. of RMW rises from 2.1% to 2.6%, and its  $t$  falls from 2.5 to 2.1
- At the  $t > 3$  hurdle of Section 8, HML is borderline under every choice (2.45–3.00)

## Estimated Risk Premia



- 25 size × B/M portfolios, 1963–2026; bars: 95% intervals with Shanken errors; crosses: factor means
- CAPM: intercept 14.0% (se 4.4%), market premium −4.4% (se 4.8%)
- FF3: SMB 1.4% (1.4%), HML 4.0% (1.3%); FF5: RMW 5.3% (2.1%), CMA −1.4% (2.2%)

- For a traded factor,  $\lambda$  should equal its time-series mean: HML 4.0% vs 3.6%, close; market (FF3) −6.5% vs 7.2%, far
- Market betas span only 0.86–1.42 (cross-sectional s.d. 0.14):  $\lambda_m$  is **weakly identified**, and a large intercept absorbs the level (Lewellen, Nagel & Shanken (2010))

## Useless and Weak Factors

- A **useless factor**  $g_t$  is independent of all returns:  $\beta_g = 0$ , so  $\hat{\beta}_{i,g} = O_p(T^{-1/2})$  is pure noise
- Pass 2 divides by this noise:  $\hat{\lambda}_g$  does not converge to zero and its  $t$ -test over-rejects when the model is misspecified ([Kan & Zhang \(1999\)](#))
- Simulation on the 25 portfolios, CAPM plus an independent Normal factor, 2000 replications, nominal level 5%
  - ▶  $\lambda_g = 0$  rejected in 60% (Fama–MacBeth errors) and 27% (Shanken errors) of the replications
  - ▶ First-pass Wald test of  $\beta_{i,g} = 0$  for all  $i$ : rejected in 6.7%; run it before pass 2
- **Weak factors** (little dispersion in betas, like the market here): identification-robust tests ([Kleibergen \(2009\)](#)); with many assets, consistency of pass 2 depends on factor strength ([Anatolyev & Mikusheva \(2022\)](#))

## A Skeptical Appraisal: Adding 30 Industry Portfolios

| Model | OLS $R^2$ |         | GLS $R^2$ |         |
|-------|-----------|---------|-----------|---------|
|       | 25        | 25 + 30 | 25        | 25 + 30 |
| CAPM  | 0.08      | 0.00    | 0.06      | 0.01    |
| FF3   | 0.66      | 0.22    | 0.21      | 0.08    |
| FF5   | 0.76      | 0.24    | 0.26      | 0.10    |

Monthly 1963–2026, free intercept and premia

- The 25 portfolios have a strong three-factor structure: any factors correlated with SMB and HML fit their means, so a high OLS  $R^2$  is a low hurdle ([Lewellen, Nagel & Shanken \(2010\)](#))
- Three checks
  - ▶ Broaden the test assets: FF3 falls from 0.66 to 0.22 once 30 industries are added
  - ▶ Report the **GLS** (Generalised Least Squares)  $R^2$ , weighted by  $\hat{V}^{-1}$  like GRS: never above 0.26
  - ▶ Impose  $\lambda = \mathbb{E}[f]$  for traded factors, no intercept: FF3 on the 55 assets gives  $R^2 = -0.27$
- Stocks instead of portfolios: more dispersion in betas, larger errors in variables ([Ang, Liu & Schwarz \(2020\)](#))

## Factors as a Stochastic Discount Factor



Lars Peter Hansen, Nobel Prize in  
Economics 2013

Photo: Bengt Nyman (2013); CC BY  
2.0; Wikimedia Commons

- **SDF** (Stochastic Discount Factor)  $m_t$ : prices all excess returns,  $\mathbb{E}[m_t R_{i,t}^e] = 0$
- A linear factor model is an SDF,  $m_t = 1 - \mathbf{b}^\top (\mathbf{f}_t - \mathbb{E}\mathbf{f})$
- Substitute:  $\mathbb{E}[R_i^e] = \text{Cov}(R_i^e, \mathbf{f}_t)^\top \mathbf{b} = \beta_i^\top \boldsymbol{\lambda}$  with  $\boldsymbol{\lambda} = \boldsymbol{\Sigma}_f \mathbf{b}$ : the beta pricing model of Section 6
- $\mathbf{b} \neq 0$  asks whether a factor helps price the assets;  $\boldsymbol{\lambda} \neq 0$  whether it is rewarded
- These moment conditions are estimated by GMM, [Hansen \(1982\)](#); survey of the discount-rate view: [Cochrane \(2011\)](#)

## Comparing Models: Maximum Sharpe Ratios

| Max. Sharpe ratio (annual) | CAPM | FF3  | Carhart | FF5  | FF5 + MOM |
|----------------------------|------|------|---------|------|-----------|
| 1963–2026 ( $T = 757$ )    | 0.47 | 0.66 | 0.97    | 0.97 | 1.16      |
| 2000–2026 ( $T = 319$ )    | 0.48 | 0.54 | 0.67    | 0.97 | 1.02      |

- Barillas & Shanken (2018): judge a model by  $SR^2(f) = \mu_f^\top \Sigma_f^{-1} \mu_f$  of its factors; the test assets do not matter for the comparison
- Nested models: in-sample  $SR^2$  can only rise, so test the alpha of the added factor on the others (spanning regression)
  - ▶ HML on the other five factors:  $\alpha = 0.9\%$  a year ( $t = 0.75$ ); since 2000  $-1.7\%$  ( $t = -0.81$ ): no significant incremental contribution of HML, as in Fama & French (2015)
- Non-nested FF5 vs Carhart: difference in annual  $SR^2$  0.00, 95% interval  $[-0.57, 0.66]$ ; since 2000 0.50,  $[-0.24, 1.48]$  (stationary bootstrap, Politis & Romano (1994))
- Another model-independent metric, the Hansen–Jagannathan distance: Appendix

## The Factor Zoo

- ▣ Cochrane (2011) called the growing list of new factors a “zoo”
- ▣ Harvey, Liu & Zhu (2016) catalogue hundreds of published factors (316 in their list)
- ▣ Each was tested as if it were the only one:  $|t| > 1.96$  at 5%
- ▣ With many tests, some pass by chance: **data mining**, also called p-hacking
- ▣ Their proposal: a new factor needs  $|t| > 3.0$

## Controlling False Discoveries

- **FDR** (False Discovery Rate): expected share of false rejections among all rejections; **FWER** (Family-Wise Error Rate): probability of at least one false rejection (Chapter 2)
- **BH procedure** (Benjamini & Hochberg (1995)): find the largest  $j$  with  $p_{(j)} \leq \frac{j}{M} \alpha$  and reject  $H_{(1)}, \dots, H_{(j)}$

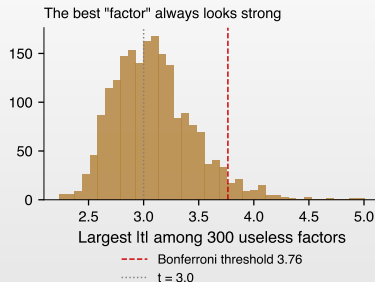
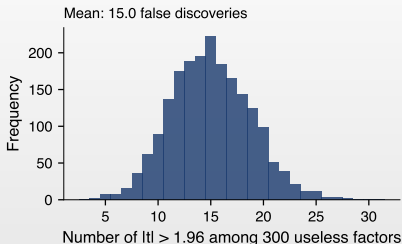
| Procedure                         | Controls | Reject $H_{(j)}$ if                                             | Valid under                 |
|-----------------------------------|----------|-----------------------------------------------------------------|-----------------------------|
| Bonferroni                        | FWER     | $p_{(j)} \leq \alpha/M$                                         | any dependence              |
| Holm (Holm (1979))                | FWER     | $p_{(k)} \leq \alpha/(M - k + 1)$ for all $k \leq j$            | any dependence              |
| BH (Benjamini & Hochberg (1995))  | FDR      | $j \leq \text{largest } k \text{ with } p_{(k)} \leq k\alpha/M$ | independence, positive dep. |
| BY (Benjamini & Yekutieli (2001)) | FDR      | as BH, thresholds divided by $\sum_{k=1}^M 1/k$                 | any dependence              |

- Positive dependence: the test statistics of the true nulls move up together (the Benjamini–Yekutieli PRDS condition: positive regression dependence on a subset); FDR control suits a search over many candidate factors

## Worked Example: Ten Candidate Factors

- Sorted  $p$ -values: 0.0004, 0.002, 0.006, 0.011, 0.018, 0.027, 0.041, 0.070, 0.15, 0.62;  $\alpha = 5\%$
- Naive ( $p \leq 0.05$ ): **7** discoveries
- Bonferroni ( $p \leq 0.005$ ): **2**
- Holm:  $0.0004 \leq 0.005$ ,  $0.002 \leq 0.0056$ ,  $0.006 \leq 0.00625$ ,  $0.011 > 0.0071$  stop: **3**
- BH thresholds  $0.005j$ :  $p_{(6)} = 0.027 \leq 0.030$  is the last success: **6**
- Seminar 3, problem A5 repeats the procedure on a different vector, with a threshold chart

## Simulation: 300 Useless Factors



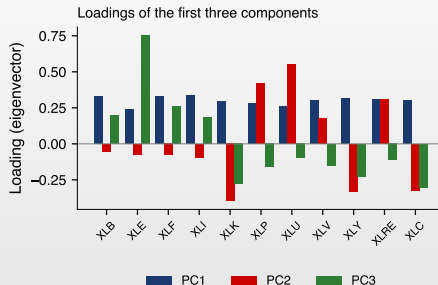
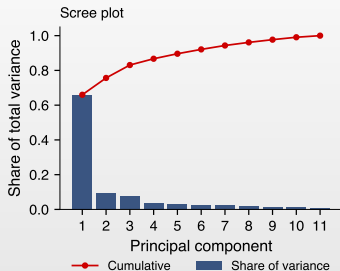
- 2000 replications; each draws 300 independent  $t$ -statistics from the standard Normal distribution (no true premium)
- On average 15.0 useless factors pass  $|t| > 1.96$ ; expected value  $300 \times 0.05 = 15$

- On average 0.83 pass  $|t| > 3$ ; at least one passes in 57% of the replications; median largest  $|t|$ : 3.06
- Bonferroni threshold for 300 tests:  $|t| > 3.76$ ; hurdles rise with the number of tests actually tried, including unpublished ones

## Principal Components in Brief

- ▣ **Latent factors**: estimated from returns alone, without naming them in advance
- ▣ **PCA** (Principal Component Analysis) of the correlation matrix of standardised returns:  $C = \Gamma \Lambda \Gamma^\top$ , components  $y_t = \Gamma^\top z_t$
- ▣ Eigenvalues  $\lambda_1 \geq \dots \geq \lambda_N$ ; eigenvectors are the **loadings**; component  $k$  explains  $\lambda_k/N$  of the variance
- ▣ Two assets with correlation  $\rho$ :  $\lambda = 1 \pm \rho$ , eigenvectors  $(1, 1)/\sqrt{2}$  (the “market”) and  $(1, -1)/\sqrt{2}$  (the spread); in a crisis  $\rho$  rises and PC1 dominates
- ▣ Textbook: [Härdle & Simar \(2019\)](#), chapter on principal components;  $N$  equicorrelated assets: Seminar 3, problem A8

## PCA of the US Sector ETFs



- 11 sector ETFs, daily log returns, 20 June 2018 – 18 September 2026 ( $T = 2073$  common days)
- PC1 explains 66.0%; all loadings positive (0.24–0.34); correlation with SPY 0.954: strongly market-related

- PC2 (9.7%): utilities +0.55, staples +0.42, real estate +0.31 vs technology -0.40, consumer discretionary -0.33: **defensive versus growth**, with no name given in advance
- PC3 (7.4%) is dominated by energy; three components explain 83.1%

## How Many Factors?

- Approximate factor model  $x_{it} = \lambda_i^\top F_t + e_{it}$ ,  $N, T \rightarrow \infty$ ;  $V(k)$ : mean squared residual after the first  $k$  components
- Bai & Ng (2002):  $\hat{k} = \arg \min_k \left[ \ln V(k) + k \frac{N+T}{NT} \ln \min(N, T) \right]$  (criterion  $IC_{p2}$ ); Ahn & Horenstein (2013): eigenvalue ratio  $\hat{k} = \arg \max_k \mu_k / \mu_{k+1}$ , no penalty to choose
- Kaiser ( $\lambda_k > 1$ ) has no asymptotic justification: the noise eigenvalues of a correlation matrix scatter around 1

| Panel                                               | $N$ | $T$  | Kaiser | Bai–Ng | Ahn–Horenstein |
|-----------------------------------------------------|-----|------|--------|--------|----------------|
| 11 sector ETFs, daily 2018–2026                     | 11  | 2073 | 2      | 5      | 1              |
| 100 size $\times$ B/M portfolios, monthly 1963–2026 | 95  | 757  | 3      | 4      | 1              |

- Standardised returns; maximum  $k$ : 5 for sectors, 8 for portfolios; 5 portfolios with missing months are dropped
- Ahn–Horenstein picks the dominant market factor (ratio 14.6); its next peak is at  $k = 3$  (ratio 3.5), the FF3 structure
- Bai–Ng on 11 sectors hits the upper bound  $k = 5$ : the penalty is built for large  $N$ , so with small  $N$  the answer is fragile

## Limits of PCA for Asset Pricing

- ▣ Loadings are **constant**: PCA cannot follow a stock whose size or value changes
- ▣ Components explain **variance**, not necessarily **average returns**
- ▣ Individual stocks enter and exit the sample: a stock panel is unbalanced
- ▣ Component signs are arbitrary; rotations preserve principal components only within repeated-eigenvalue eigenspaces, while a latent-factor model is identified only up to rotation
- ▣ Risk-premium PCA and three-pass premia keep the PCA backbone but target pricing (Appendix); PCA and factor analysis in finance: [Franke, Härdle & Hafner \(2019\)](#)

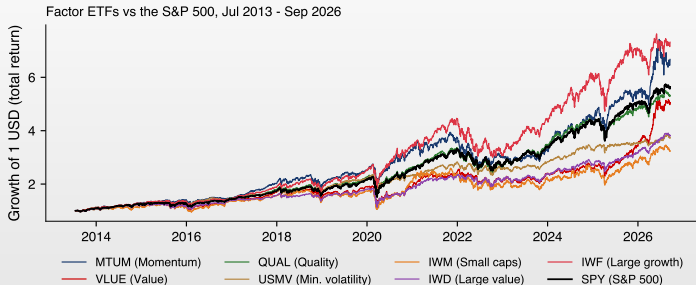
## Instrumented PCA

- ▣ **IPCA:** Instrumented Principal Component Analysis, [Kelly, Pruitt & Su \(2019\)](#)
- ▣ Loadings are functions of  $L$  observed characteristics  $\mathbf{z}_{i,t}$  (size, B/M, momentum, ...)

$$R_{i,t+1}^e = \alpha_{i,t} + \boldsymbol{\beta}_{i,t}^\top \mathbf{f}_{t+1} + \varepsilon_{i,t+1}, \quad \boldsymbol{\beta}_{i,t}^\top = \mathbf{z}_{i,t}^\top \boldsymbol{\Gamma}_\beta$$

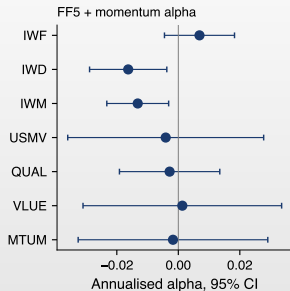
- ▣  $\boldsymbol{\Gamma}_\beta$  ( $L \times K$ ) and the latent factors  $\mathbf{f}_t$  are estimated jointly by alternating least squares
- ▣ Test: are characteristics priced through betas ("covariances"), or directly as alphas ( $\alpha_{i,t} = \mathbf{z}_{i,t}^\top \boldsymbol{\Gamma}_\alpha$ )?
- ▣ Result in their US stock data: a few IPCA factors price the cross section better than observable-factor models with as many factors, and  $\boldsymbol{\Gamma}_\alpha$  is insignificant: "characteristics are covariances"

## Factor ETFs versus the Market



- **Smart beta:** rule-based long-only portfolios that tilt towards factors, sold as ETFs; they keep market exposure and need not replicate an academic long-short factor
- Growth of 1 USD with dividends re-invested, from 18 July 2013 (first common day)
- **CAGR:** large growth (IWF) 16.3% and momentum (MTUM) 15.5% beat SPY (14.0%); quality (QUAL) 13.5%, value (VLUE) 13.1%, large value (IWD) 10.7%, minimum volatility (USMV) 10.5%, small caps (IWM, Russell 2000 index) 9.3%
- Thirteen years of one realisation: not evidence against the premia on its own

## Alphas of Factor ETFs



Factor loadings

|      | Mkt-RF | SMB   | HML   | RMW   | CMA   | MOM   |
|------|--------|-------|-------|-------|-------|-------|
| MTUM | 1.03   | -0.12 | 0.01  | -0.22 | -0.05 | 0.36  |
| VLUE | 0.98   | 0.08  | 0.34  | -0.06 | 0.09  | -0.03 |
| QUAL | 0.97   | -0.08 | -0.04 | 0.15  | 0.03  | -0.04 |
| USMV | 0.74   | -0.11 | -0.02 | 0.20  | 0.21  | -0.02 |
| IWM  | 1.04   | 0.85  | 0.08  | -0.10 | -0.02 | 0.03  |
| IWD  | 0.94   | -0.04 | 0.29  | 0.09  | 0.19  | -0.04 |
| IWF  | 1.03   | -0.11 | -0.28 | -0.03 | -0.12 | -0.00 |

- Daily regressions on FF5 + MOM, Newey–West errors, each ETF on its own trading calendar (2000–2026)
- The loadings match the labels: MTUM on MOM 0.36, VLUE on HML 0.34, IWM on SMB 0.85; IWF -0.28 on HML; USMV market beta 0.74

- Alphas close to zero: MTUM -0.2% ( $t = -0.11$ ), QUAL -0.3%; Russell ETFs IWM -1.3% ( $t = -2.57$ ), IWD -1.6% ( $t = -2.55$ ), consistent with costs and index construction
- $R^2$  between 0.81 and 0.97: mostly repackaged factor exposure

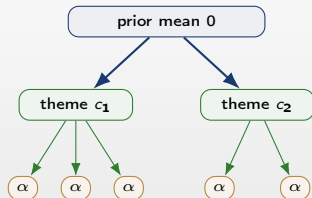
## Timing, Crowding and Costs

- **Factor timing:** switching between factors based on signals (valuation spreads, recent returns)
  - ▶ Noisy premia and short histories make timing hard to verify out of sample (Chapter 8)
- **Crowding:** many investors hold the same factor positions
  - ▶ Premia can shrink ([McLean & Pontiff \(2016\)](#)), and crowded positions can amplify losses during forced unwinding; the 2009 momentum crash coincided with a sharp rebound of past losers ([Daniel & Moskowitz \(2016\)](#))
- **Implementability:** long–short factors ignore trading costs and short-selling fees; ETFs pay them

## Case Study: Research Question

- **The challenge** to factor research takes two forms (Jensen, Kelly & Pedersen (2023), Introduction)
  - ▶ No internal validity: results cannot be reproduced; Hou, Xue & Zhang (2020) replicate only 35% of their anomalies
  - ▶ No external validity: results come from testing many factors without controlling the FDR; Harvey, Liu & Zhu (2016) conclude that most claimed findings are likely false
- **Question:** do published factors have significant CAPM alphas in the original data, out of sample and in other countries, once multiple testing is taken into account?
- **Why it matters:** if most factors are false discoveries, multifactor models and factor investing (Sections 5 and 10) rest on noise
- **A second view of the zoo:** most new factors are redundant given the existing ones (Feng, Giglio & Xiu (2020), Appendix); the paper instead treats the zoo as correlated evidence on a few economic themes
- **Link to this chapter:** Jensen's alpha (Section 4) is the object tested; Benjamini–Yekutieli (Section 8) is the frequentist benchmark; Bayesian shrinkage extends the Vasicek beta (Section 3) to alphas

## Case Study: Data and Method



each factor's alpha borrows strength from the other factors of its theme

- **Data** (Section II): 153 factors in 93 countries; terciles of each characteristic, capped value weights, long the side with the higher expected return in the original paper; each factor scaled to a 10% annual idiosyncratic volatility
- **Themes**: 13 clusters of US factors, hierarchical clustering on the correlation of CAPM residuals
- **Model** (Section I):  $f_{i,t} = \alpha_i + \beta_i r_t^m + \varepsilon_{i,t}$ , alphas  $\alpha_i = c_j + s_n + w_i$  (theme, signal, factor), prior mean 0
- **One factor**:  $\mathbb{E}(\alpha \mid \hat{\alpha}) = \kappa \hat{\alpha}$ ,  $\kappa = 1/(1 + \sigma^2/(\tau^2 T))$  (eq. 5): short, noisy samples are shrunk more, as in Vasicek
- **Replication rate**: share of factors with a significant alpha; Bayesian discovery if posterior mean / posterior s.d.  $\geq 1.96$
- **Full hierarchy and estimation**: Appendix

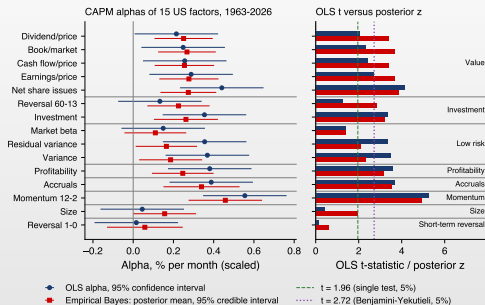
## Case Study: Key Results

### Replication rates, US (Figure 1)

|                                      |       |
|--------------------------------------|-------|
| Hou, Xue & Zhang (2020), raw returns | 35.0% |
| This paper, raw returns              | 55.6% |
| CAPM alpha, OLS                      | 82.4% |
| CAPM alpha, Benjamini–Yekutieli      | 75.6% |
| CAPM alpha, empirical Bayes          | 82.4% |
| World, empirical Bayes               | 82.4% |

- Testing alphas instead of raw returns drives most of the gap with Hou, Xue & Zhang (2020)
- Bayesian FDR 0.1%: shrinkage controls false discoveries without the power loss of a higher  $t$  hurdle
- **Decay**: mean alpha 0.49% a month in sample, 0.26% after the original sample (−47%); slope of post-sample on in-sample alphas 0.26 (Figure 8), against about 0.9 from shrinkage alone
- The strongest factors decay more: arbitrage or data mining (Section 12)
- Complete results: Appendix

## Case Study: Replication on Course Data



- 15 US long-short factors from value-weighted sorts of the [Kenneth French Data Library](#), July 1963 – July 2026 ( $T = 757$ ); CAPM alphas scaled to 10% idiosyncratic volatility; themes of the paper ([authors' cluster labels](#))
- Replication rates: OLS 73.3% (11 of 15), Benjamini-Yekutieli 53.3% (8; critical  $t = 2.72$ ), empirical Bayes 86.7% (13)

- Shrinkage towards the theme, not only towards zero: long-term reversal (investment theme)  $0.13\% \rightarrow 0.23\%$  a month ( $t = 1.25$ , posterior  $z = 2.85$ )
- Failures: market beta and short-term reversal ( $z \leq 1.41$ ); size only just passes ( $z = 1.96$ ), echoing the paper, where size is one of the two themes below 50%

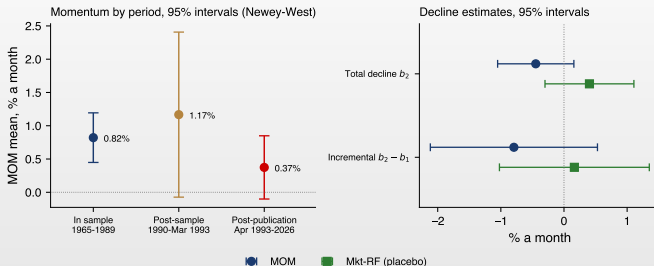
## Case Study: Limits and an Open Question

- ▣ **Limits of the course version**
  - ▶ 15 US signals instead of 153 in 93 countries; Kenneth French Data Library sort breakpoints instead of non-micro terciles with capped weights; one region, so no signal level ( $\tau_s = 0$ )
  - ▶ The rate depends on the theme structure: with a single common theme the empirical-Bayes rate is 100%
- ▣ **Limits stated by the authors:** CAPM as the only benchmark, uniform construction of the characteristics, unobserved factors in the prior (Appendix)
- ▣ **Open question:** the paper's data end in 2020; did the posterior alphas and the replication rate hold in 2021–2026, and can a prior estimated on US factors sharpen short-sample BVB factors?
- ▣ Data and code of the paper: [jkpfactors.com](https://jkpfactors.com); course project: Seminar 3, C3

## An Open Question and the Discovery Loop

- **Open question:** why do factor premia decay after publication: arbitrage, data mining or a change in risk, and does it differ across factors and markets?
- Evidence so far
  - ▶ [McLean & Pontiff \(2016\)](#): across 97 predictors, returns are 26% lower out of sample and 58% lower after publication; the gap is attributed to publication-informed trading
  - ▶ [Jensen, Kelly & Pedersen \(2023\)](#): most factors replicate, but the strongest decay most; [Chen & Zimmermann \(2022\)](#): open data and code for the published predictors
- **One pass of the loop**
  - ▶ The assistant drafts: a literature table with DOIs, competing hypotheses with their predicted patterns, the regression code, a robustness plan
  - ▶ The human fixes before running: the event dates (Jegadeesh–Titman: sample ends 1989, published March 1993), the factor set, the multiple-testing correction, the placebo
  - ▶ Checks: every DOI resolves; the chapter's split-sample  $t$ -statistics are reproduced first; the data vintage is recorded
- Reusable prompts and the full checklist: Appendix

## Mini-Case: Was Momentum Arbitrated Away?



- Design of McLean & Pontiff (2016):  

$$MOM_t = a + b_1 \text{PostSample}_t + b_2 \text{PostPub}_t + e_t$$
, monthly 1965–2026, Newey–West errors; post-sample 1990-01 – 1993-03, post-publication from 1993-04
- $b_2$ : total decline relative to the in-sample mean;  $b_2 - b_1$ : incremental effect of publication

- $a = 0.82\%$  a month ( $t = 4.32$ );  $b_2 = -0.45\%$  ( $t = -1.45$ , interval  $[-1.05\%, 0.16\%]$ ): 54% lower;  $b_2 - b_1 = -0.79\%$  ( $t = -1.18$ ); placebo Mkt-RF  $b_2 = +0.40\%$  ( $t = 1.12$ )
- Close to the 58% of McLean & Pontiff (2016), but one factor cannot reject  $H_0 : b_2 = 0$  at 5%; arbitrage is one interpretation, not an identified cause

## Your Turn: A Project Seed

- **Question:** do the Fama–French factors and momentum decay after publication, and does the decay differ across factors?
  - ▶ Pooled regression of SMB, HML, RMW, CMA and MOM on factor-specific post-sample and post-publication dummies; Mkt-RF as placebo; errors robust to cross-factor correlation and autocorrelation
  - ▶ Whether decay is larger where arbitrage is easier needs stock-level proxies (size, liquidity, short-sale costs), not factor returns
- **Replicate first:** the split-sample  $t$ -statistics and the mini-case estimates
- **Deliverables:** a pre-registration note ( $H_0$ ,  $H_1$ , dates from each paper, factor set, corrections) committed before the results; code as a Quantlet; a report with Newey–West inference, placebo periods and power
- **Logs:** prompts in `AI_USE.md`; every caught error (a wrong publication date, a misquoted number) in `AI_ERRORS.md`

## Key Takeaways: An Empirical Workflow

1. **Estimate exposures** with HAC errors, report sample and proxy, shrink extreme betas (Blume, Vasicek)
  2. **Test the pricing restriction jointly**: GRS measures the Sharpe-ratio gain from the test assets; the CAPM fails on the 25 size  $\times$  B/M portfolios (flat SML,  $GRS = 4.20$ )
  3. **Estimate premia** in two passes with errors robust to estimated betas and misspecification; a high  $R^2$  on 25 portfolios is a low hurdle
  4. **Guard against the zoo**: multiple-testing hurdles, spanning tests, model comparison by maximum Sharpe ratios
  5. **Replicate**: latent factors and out-of-sample evidence; factor ETFs are cheap factor exposure, not alpha
- Open: why premia decay after publication; how to price with time-varying betas; which factors price assets in small markets such as the BVB

## Check Yourself, and Next

### Retrieval questions

- Why does the HAC standard error of a beta differ from the classical one, while the beta does not?
- Why can GRS fail to reject the CAPM on sector ETFs but reject it on the 25 portfolios over the same months?
- Why does a useless factor look priced in pass 2?
- Why can the empirical-Bayes replication rate exceed the Benjamini–Yekutieli rate?

### Links to other chapters

- Chapter 2: an alpha is an anomaly or a missing risk factor, the joint hypothesis problem
- Chapter 4: factor covariance models and shrinkage for Markowitz
- Chapters 5–6: time-varying volatility, correlation and betas
- Chapter 8: backtesting factor strategies without overfitting
- Chapter 13: machine learning for the cross section of returns

## Appendix: GRS under the Null Hypothesis

- Under  $H_0$  and the exact assumptions,  $W \sim F(N, T - N - 1)$ , so  $\mathbb{E}[W] = \frac{T-N-1}{T-N-3}$
- Invert the Sharpe-ratio form of  $W$ : the squared Sharpe-ratio gain expected under  $H_0$  is

$$\mathbb{E}[\widehat{SR}^2(f, R) - \widehat{SR}^2(f)] \approx \frac{N(1 + \widehat{SR}^2(f))}{T - N - 3}$$

- 25 portfolios,  $T = 757$ :  $25 \times 1.018/729 = 0.035$  (monthly), against  $0.405^2 - 0.134^2 = 0.146$  observed
- In-sample maximisation over  $N$  assets always adds some Sharpe ratio: the test asks whether the gain exceeds this noise level
- 9 sector ETFs,  $T = 331$ : observed gain  $0.191^2 - 0.146^2 = 0.015$ , below the expected  $9 \times 1.021/319 = 0.029$

## Appendix: Tests with Many Assets

- $\hat{\Sigma} = \hat{E}^\top \hat{E} / T$  has rank at most  $T - 2$  (two parameters per regression)
  - ▶ Example: 100 size  $\times$  B/M portfolios on 8 years of monthly data ( $T = 96$ ):  $\hat{\Sigma}$  is singular, GRS does not exist
  - ▶ Well before this point the inverse is badly estimated and power falls
- Tests for many assets that avoid  $\hat{\Sigma}^{-1}$ : standardised sums of squared  $t$ -statistics, [Pesaran & Yamagata \(2024\)](#)
- Power enhancement by screening large alphas: [Fan, Liao & Yao \(2015\)](#)
- With many assets, consistency of the second pass depends on factor strength: [Anatolyev & Mikusheva \(2022\)](#)

## Appendix: Misspecification-Robust Standard Errors

- Fama–MacBeth and Shanken errors assume a **correct** model: all pricing errors  $e = \mu - X\gamma$  are zero ( $X = [1, B]$ )
- GRS rejects here, so  $\gamma$  is a pseudo-true value; the noise in  $\hat{B}$  multiplies  $e$  and adds a variance term (Kan, Robotti & Shanken (2013))
- Write both passes as one exactly identified GMM system: moments  $(R_t - a - Bf_t) \otimes (1, f_t)$  and  $X^\top (R_t - X\gamma)$ ; the sandwich variance with the full Jacobian is the robust one
  - ▶ Setting  $e = 0$  in the Jacobian gives the correct-model, heteroskedasticity-robust version of Jagannathan & Wang (1998)
- FF5 on the 25 portfolios: s.e. of RMW 2.1%  $\rightarrow$  2.6%, CMA 2.2%  $\rightarrow$  2.7%, intercept 3.3%  $\rightarrow$  3.8%
- RMW:  $t$  falls from 2.5 to 2.1; HML changes by less than 0.05 percentage points
- Advanced course:  Generalized Method of Moments

## Appendix: Hansen–Jagannathan Distance and Bayesian Selection

- **Hansen–Jagannathan distance** (Hansen & Jagannathan (1997)): least-squares distance of the model SDF from the set of valid SDFs

$$\delta^2 = \min_b e(b)^\top \mathbb{E}[R^e R^{e\top}]^{-1} e(b), \quad e(b) = \mathbb{E}[m_t(b) R_t^e]$$

- ▶ Excess returns  $R_t^e$  with the normalisation  $\mathbb{E}[m] = 1$ ; for gross returns  $R_t$ :  $e = \mathbb{E}[mR] - 1$ , and the SDF needs a free scale ( $\mathbb{E}[m] = 1/(1 + R_f)$ )
- ▶ Same weighting matrix for every model, so distances are comparable; the test statistic has a weighted  $\chi^2$  limit
- Zoo-scale selection: Bayesian averaging over two quadrillion factor models, robust to weak factors (Bryzgalova, Huang & Julliard (2023))
- SDF shrinkage on characteristic-based factors: a few principal components of anomaly returns capture the SDF (Kozak, Nagel & Santosh (2020))
- Common lesson: compare models by a single, model-independent metric, with its sampling uncertainty

## Appendix: Taming the Zoo with Double-Selection LASSO

- Feng, Giglio & Xiu (2020): does a new factor add information given the existing zoo?
- **Double-selection LASSO** (Least Absolute Shrinkage and Selection Operator)
  - ▶ Step 1: a LASSO of average test-asset returns on their covariances with hundreds of factors selects the controls that price the cross section
  - ▶ Step 2: a second LASSO selects the factors whose covariances explain those of the new factor, so that omitted controls do not bias its SDF coefficient
  - ▶ Step 3: the SDF coefficient of the new factor is estimated with the union of both sets of controls and tested
- Result: most new factors are redundant; only a few add pricing information
- The case study reaches a compatible conclusion: the zoo is highly redundant, and a few themes carry the information

## Appendix: More on the Number of Factors

- ▣ Onatski (2010): separates factor eigenvalues from the edge of the noise eigenvalue distribution
- ▣ Kaiser ( $\lambda_k > 1$ ) on a correlation matrix: with  $N$  assets and pure noise the eigenvalues spread around 1 with a width that depends on  $N/T$ , so the rule has no asymptotic justification
- ▣ Bai–Ng penalties are built for large  $N$  and  $T$ : with 11 sectors the criterion keeps falling up to  $k_{\max}$
- ▣ Ahn–Horenstein needs no penalty: the ratio of consecutive eigenvalues peaks after the last strong factor
- ▣ Report the range of estimates and the sensitivity to  $k_{\max}$ , not a single number (Seminar 3, problem B3)

## Appendix: Risk-Premium PCA and Three-Pass Premia

- ▣ **RP-PCA** (Risk-Premium PCA, [Lettau & Pelger \(2020\)](#)): principal components of  $\frac{1}{T}X^T X + \gamma \bar{x}\bar{x}^T$ 
  - ▶  $\gamma = -1$  gives covariance PCA;  $\gamma > -1$  also rewards factors that explain average returns
  - ▶ In their data it finds factors with higher Sharpe ratios and smaller pricing errors than PCA
- ▣ **Three-pass estimator** ([Giglio & Xiu \(2021\)](#)): the premium of an observed factor  $g_t$  when the model omits factors
  - ▶ (1) PCA of test-asset returns; (2) cross-sectional regression of means on the latent betas; (3) regress  $g_t$  on the latent factors and map their premia to  $g_t$
- ▣ Both keep PCA's statistical backbone but target pricing, not only variance; IPCA adds characteristics

## Appendix: IPCA and Crypto-Asset Factors

- Title of the IPCA paper, [Kelly, Pruitt & Su \(2019\)](#): “Characteristics are covariances”
- In their US stock data, a few IPCA factors price the cross section better than observable-factor models with as many factors
- The alpha term  $\Gamma_\alpha$  becomes insignificant: characteristics predict returns because they measure risk exposure
- IPCA links PCA (statistics) with characteristic sorts (Fama–French): time-varying betas, few factors
- Crypto-assets: market, size and momentum factors price the cross section of coins ([Liu, Tsyvinski & Wu \(2022\)](#)); dimension reduction on crypto and traditional return features: [Pele et al. \(2023\)](#), [Jheng et al. \(2026\)](#)
- Advanced course:  [Multivariate Statistical Analysis](#)

## Appendix: Case Study, Data and Construction

- **Data** (Section II): 153 factors in 93 countries; 138 from the list of [Hou, Xue & Zhang \(2020\)](#) plus 15 from the wider literature; US returns from 1926, most developed markets from 1986; all in US dollars, in excess of the US Treasury bill
  - ▶ Each country and month: terciles of the characteristic, breakpoints from non-micro stocks; “capped” value weights (market equity winsorised at the NYSE 80th percentile); long the tercile with the higher expected return in the original paper, one-month holding
  - ▶ At least 5 stocks per leg and 60 monthly observations; every factor scaled to a 10% annual idiosyncratic volatility
  - ▶ 13 themes: hierarchical clustering of US factors, distance 1– correlation of CAPM residuals, Ward linkage (Section II.B)
- **Model** (Section I):  $f_{i,t} = \alpha_i + \beta_i r_t^m + \varepsilon_{i,t}$ ; alphas  $\alpha_i = c_j + s_n + w_i$  (theme  $j$ , signal  $n$ , factor  $i$ ; eq. 21), prior mean 0
  - ▶ Prior variance  $\Omega = MM^\top \tau_c^2 + ZZ^\top \tau_s^2 + I\tau_w^2$  (eq. 23); posterior mean  $(\Omega^{-1} + T\Sigma^{-1})^{-1} T\Sigma^{-1} \hat{\alpha}$  (Proposition 4)
  - ▶ Empirical Bayes:  $\tau_c, \tau_s, \tau_w$  by maximum likelihood of  $\hat{\alpha} \sim N(0, \Omega + \hat{\Sigma}/T)$  (Section I.C.2)
- **Replication rate**: share of factors significant in the original paper whose alpha is significant here; Bayesian discovery if posterior mean / posterior s.d.  $\geq 1.96$  (eq. 26)

## Appendix: Case Study, Complete Results

- Replication rates (Figure 1): [Hou, Xue & Zhang \(2020\)](#) 35.0%; US raw return 55.6%; originally significant 61.3%; CAPM alpha 82.4%; Benjamini–Yekutieli 75.6%; empirical Bayes 82.4%; world, empirical Bayes 82.4%
- OLS: 98 of 119 US factors; Benjamini–Yekutieli needs  $|t| > 2.7$  (Figure 4)
- Bayesian FDR 0.1%, 95% interval [0, 1%]; expected share of true factors 94% (Section III.C)
- Mega and large stocks: 77.3% and 79.8% (Figure 5); 11 of 13 themes at least 50%, all but low leverage and size: not an artefact of micro-caps
- Mean alpha 0.49% a month in sample, 0.26% after the original sample:  $-47\%$ ; slope of post-sample alphas on in-sample alphas 0.26 (Figure 8), against about 0.9 expected from shrinkage alone
- Tangency Sharpe ratio (monthly), US 1972–2020: market 0.13, all factors 0.80 (Figure 14)

## Appendix: Case Study, the Course Replication in Detail

- Benjamini–Yekutieli is the strictest, as in the US rates of the paper: 82.4%, 75.6%, 82.4% (Figure 1)
- Net share issues move towards their theme: 0.44%  $\rightarrow$  0.27% a month
- Posterior s.d. 0.07%–0.10% against OLS s.e. 0.11%: each theme pools the evidence of its members
- **Prior dispersion:**  $\hat{\tau}_c = 0.28\%$ ,  $\hat{\tau}_w = 0.04\%$  a month, against 0.35% and 0.21% in Table I; 15 signals leave little factor-specific dispersion
- **Themes:** the 15 signals fall into 8 of the paper's 13 themes (5 in value, 3 in low risk, 2 in investment); with a single common theme the empirical-Bayes rate is 100%

## Appendix: Case Study, Limits Stated by the Authors and Extensions

### ▣ **Limits stated by the authors**

- ▶ CAPM is the only benchmark: low-leverage factors fail on CAPM alpha but matter jointly with other themes (Section III.A)
- ▶ Characteristics are built consistently, sometimes unlike the original paper (Section II.C)
- ▶ Unobserved factors bias the prior: with the scenarios of [Harvey, Liu & Zhu \(2016\)](#) the US rate falls to 81.5% and 79.8% (Figure 10)

### ▣ **Complete the course replication** (Seminar 3, C3): justify each sign from the original paper, compare the dendrogram with the paper's themes, add the Bayesian FDR and the post-sample slope (Figure 8)

### ▣ Data and code of the paper: [jkpfactors.com](http://jkpfactors.com), [github.com/bkelly-lab/ReplicationCrisis](https://github.com/bkelly-lab/ReplicationCrisis)

## Appendix: The Discovery Loop, Reusable Prompts (1/2)

### ▣ Literature

- ▶ Summarise the evidence since 2016 on post-publication decay and replication of anomaly returns. For each paper: DOI, sample, number of factors, headline estimate, identification of the decay. Flag entries you are unsure of.
- ▶ Resolve every DOI on Crossref; check each headline number against the abstract

### ▣ Hypothesis

- ▶ Contrast data mining, publication-informed arbitrage and time-varying risk as explanations of decay. For each, describe the predicted pattern across the in-sample, post-sample and post-publication periods, and across high vs low limits-to-arbitrage stocks.

### ▣ Data and code

- ▶ Load the monthly momentum factor from the Kenneth French Data Library; from 1965, regress MOM on a post-sample dummy (1990-01 to 1993-03) and a post-publication dummy (from 1993-04), with Newey-West errors.
- ▶ Replicate a known number first: momentum  $t = 5.53$  before 2000 and 0.75 after (split-sample chart)

## Appendix: The Discovery Loop, Reusable Prompts (2/2)

### ▣ Robustness

- ▶ Write a pre-registered robustness plan: one publication date per factor, a Holm correction across factors, excluding the 2009 momentum crash, the working-paper date instead of the journal date, and 1927-1962 as a placebo period for Mkt-RF, SMB, HML and MOM (RMW and CMA start in July 1963).

### ▣ Critique

- ▶ Referee this: one factor, one publication date,  $t = -1.45$ . What is the power to detect a 58% decay, and why does pooling many anomalies change it?

### ▣ Report

- ▶ In-sample mean, both dummies with Newey–West errors, the placebo, the data vintage (download date)
- ▶ Pooled factors: errors robust to cross-factor correlation within a month and to autocorrelation
- ▶ Prompts that shaped a decision go into AI\_USE.md

- ▣ The assistant enumerates specifications; the human fixes which ones count before running them

## Appendix: What the Researcher Must Check

- **Wrong event date:** sample end, working paper and journal issue are different dates
  - ▶ test: read them from the paper itself; Jegadeesh–Titman: sample ends 1989, published March 1993
- **Choosing the factor after seeing it decay:** fix the set of factors in advance; Holm or BH across them
- **Serially correlated, crash-prone returns:** Newey–West errors; rerun without 2009
- **Data vintages:** the factor library is revised retroactively
  - ▶ test: record the download date; reproduce the chapter's split-sample  $t$ -statistics
- **Misquoted literature:** a common slip gives 58% as the out-of-sample decline; it is post-publication (26% out of sample)
- **Low power:** a single factor rarely rejects; report the confidence interval of the decay, not only its sign

## Appendix: Further Study and Code

### Advanced courses on quantinar.com

-  Generalized Method of Moments
-  Multivariate Statistical Analysis

### Code and data

- All charts and tables: [github.com/danpele/MFM/tree/main/Quantlets/Ch\\_03](https://github.com/danpele/MFM/tree/main/Quantlets/Ch_03)
- Factor data: [Kenneth French Data Library](#)

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